11–14 第2篇

阅读理解（说明文） Why Do Our Fingers Get Wrinkled After a Long Bath?

① After a long hot shower or a forty-minute soak in the bathtub, if you look carefully at the pads of your fingers, you will see **many small lines and soft hills**, like a tiny mountain range on the tips of your hands. Toes look the same way. We have all seen this since childhood; many of us still say, without thinking, "My fingers have become **pruney**" (because they look like small dried plums). For a long time people believed the cause was simple: the skin had **soaked up too much water** and swollen up. But modern research has shown that the truth is **deeper and much more interesting**.

② Let us first look at the old idea. The skin on our fingers and toes has a **thick outer layer** made of a protein called **keratin**. For years people thought that keratin, like a dry towel, simply drank the water and swelled up; the swollen skin could not stretch flat on the round fingertip and therefore **folded into wrinkles**. This explanation seems reasonable. But in 1935, a doctor named Dr. Lewis noticed something that did not fit: people whose **nerves to the fingertips had been damaged** did not get pruney fingers at all, even after a very long bath.

③ If wrinkles were just the result of soaking, it should not matter whether the nerves were healthy. But it mattered a great deal. This small observation opened a door. After many careful studies, scientists reached a surprising conclusion: **pruney fingers are not caused by the skin swelling up with water. They are caused by the body actively pulling the skin downwards — on purpose**.

④ How can the body "pull" the skin? Under the thick outer layer, there are thousands of **tiny blood vessels**. When fingers stay in warm water for a long time, the brain sends a signal along the nerves to the fingertips: **"Make the small blood vessels narrower."** The medical name for this is *vasoconstriction*. When the vessels shrink, the tissue beneath the outer skin becomes **slightly smaller**. The outer skin, which is fixed at the edges, cannot shrink at the same speed. The result is that the larger outer skin must **fold down into small ridges** — just like a tightly stretched cloth that is pushed from both sides.

⑤ So the wrinkles are, in fact, **active work** by our nervous system. But why would the body bother doing this? Here scientists have offered a beautiful idea. In 2011, a team at Newcastle University in England showed that **pruney fingers can hold wet objects more firmly** than smooth, dry ones. The tiny ridges act like the **tread on a car tyre** or the **grip on the bottom of a running shoe**: water can flow out along the small lines, so the skin touches the wet object better. The researchers showed this by asking volunteers to

move small wet glass marbles between two bowls. With pruney fingers, they were faster. With smooth fingers, slower.

⑥ This makes sense for our very distant ancestors. A human standing by a river, or a forest after heavy rain, often needed to hold **wet stones, wet roots, slippery fish, or wet tree branches**. Feet with pruney toes could also grip **wet ground** more safely. The tiny skin ridges were, quietly, a **rain-day friend** — a small extra grip that the body could switch on and off as needed. So next time your fingers look pruney after a bath, you are watching a very old piece of engineering at work.

⑦ A few interesting side-points make the story fuller. First, the wrinkles only appear in clear water. Put your fingers in **very salty water**, and they will almost never wrinkle, because the nerves do not send the same signal. Second, the wrinkle pattern on each person is **slightly different** and is even used in some countries to identify people from photographs. Third, very cold water or very hot water stops the wrinkle reflex — the body thinks something else is wrong and keeps the vessels wide. The system, it turns out, is **carefully tuned**, not accidental.

⑧ So the next time, after a long warm bath on a November evening, you step out, towel dry, and look at your pruney fingers, remember: they are not a small tired sign that your skin is getting "worn". They are a small, ancient, **active choice** your body has quietly made for you — a better grip for a rainy hand. Your body, in this small way, is still paying attention to a river it has not seen for many thousand years.

注释： pruney 像干梅子一样皱的；keratin 角蛋白；vasoconstriction 血管收缩；vessel（血）管；ridge 脊/皱纹

根据短文内容，选择最佳答案。

1. What was the old explanation for pruney fingers?

A. That the skin is broken by hot water.
B. That the thick outer layer of skin (made of keratin) drank the water and swelled up like a dry towel; the swollen skin could not stretch flat on the round fingertip, so it folded into wrinkles.
C. That the nervous system pulled the skin up.
D. That hot water changed the colour of the skin.

2. What did Dr. Lewis notice in 1935 that did not fit the old idea?

A. That fingers get wrinkled after only cold water.
B. That people whose nerves to the fingertips had been damaged did not get pruney fingers at all, even after a very long bath — which means wrinkles are not only about soaking.

C. That fingers get wrinkled on sunny days.
D. That old people do not get wrinkled fingers.

3. How does the body ACTUALLY make pruney fingers?

A. The skin absorbs water like a sponge.
B. The brain sends a signal along the nerves to the fingertips; the tiny blood vessels narrow (vasoconstriction); the tissue beneath the outer skin shrinks; the larger outer skin folds down into small ridges.
C. Air bubbles push the skin up.
D. Heat burns the skin into lines.

4. According to the 2011 Newcastle study, why are pruney fingers USEFUL?

A. They make fingers look pretty.
B. They can hold wet objects more firmly, because the tiny ridges act like the tread on a car tyre — water can flow along the small lines, so the skin touches wet objects better; volunteers moved small wet glass marbles faster with pruney fingers than with smooth fingers.
C. They make fingers longer.
D. They keep hands warm.

5. Which of the following is NOT one of the "side-points" mentioned in Paragraph ⑦?

A. In very salty water, fingers almost never wrinkle, because the nerves do not send the same signal.
B. The wrinkle pattern is slightly different for each person and can even be used for identification.
C. Very cold or very hot water stops the wrinkle reflex.
D. The wrinkle reflex only happens at night.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段旧理论叙述
【解析】吸水膨胀说直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段末 "nerves to the fingertips had been damaged"
【解析】关键反例直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段机制说明
【解析】血管收缩主动折叠机制直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段 Newcastle 研究
【解析】抓握优势实验直引。A、C、D 均错。故选 B。

5.
【答案】D
【题型】细节理解题（否定）
【定位】第⑦段三条补充
【解析】A、B、C 均在文中；D 未提及。故选 D。